Computer Network

- Introduction to CN
- Physical Layer
- Data link layer
- Network layer

70 %

- Transport layer
- Application layer

30 %

* Introduction to CN

- History / Need of CN
- N/w Models → OSI, TCP / IP
- Introduction to OSI model
- Function of each layer
- Introduction to TCP / IP model
- Introduction to IP & its supporting protocols.
- Introduction to TCP.

* Physical Layer

- Introduction to Signals
  - Analog → Period, Freq, Amplitude
  - Digital → bit, band
- Signal encoding techniques
- Bandwidth, Throughput & delay *** Numericals
- Multiplexing → FDM, WDM, TDM

Data link layer

- Error Detection vs correction
- Logic for error detection
- Hamming Distance | Minimum H.D.
- Linear block codes
  - Simple Parity check code
  - 2D "
  - CRC
  - Checksum
  - Hamming code
- Framing
- Flow & error control
  - Stop & wait
  - Go back N
  - Selection Repeat

  } Sliding Window
- Access control
  - Channelization Protocols
    - FDMA
    - TDMA
    - CDMA
  - Controlled Access protocols
    - Reservation
    - Token Passing
    - Polling
  - Random access Protocols **
    - Aloha
    - Slotted Aloha
    - CSMA
    - CSMA/CA
    - CSMA/CD **
- Ethernet
- Connecting Devices
  - Repeater & Hub
  - Bridge
  - Switch
  - Router
  - Gateway

Network Layer

- I.P addressing
  - class full
    - Subnetting
    - Supernetting
  - class less
    - Subnetting
    - Supernetting
- Special address in IPv4 Scheme
- IPv4 header
- Supporting Protocols of IP
  - ARP
  - RARP
    - BOOTP
    - DHCP
  - ICMP
  - IGMP
- Delivery, Forwarding & Routing

Transport layer

- Introduction to Transport layer
- UDP
- TCP
- Congestion control

Application layer

- N/w Security
- DNS
- Telnet
- Email
- FTP
- HTTP ***